Steven Ortiz

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Education

Rice University, Houston, TX — **GPA** 3.89
Bachelor of Science in Mechanical Engineering — **Expected May** 2028

Work Experience

Oshman Engineering Design Kitchen, Rice University

Fabrication Shop Lead, **August 2026 – Present**
Engineering Design Lab Assistant, **January 2025 – Present**

- Lead fabrication shop operations and coordinate 20 lab assistants across hands-on workshops covering CNC machining, laser cutting, waterjet cutting, 3D printing, and woodworking.
- Support students in equipment operation, troubleshooting, and safe fabrication practices across more than 40 active student projects each semester.

Cognitive Human Robotic Partnership Lab, Rice University

Undergraduate Research Assistant, **May 2026** – **August 2026**

- Characterized upper-limb movement in able-bodied and chronic stroke participants using wearable inertial sensors to study motor and compensatory movement strategies.
- Developed an iOS data-collection application in Swift and integrated MbientLab MetaMotion and LP-Research IMU sensors, building acquisition and processing tools in Python and C++ for Windows and Linux.

East End Auto Repair, East Hampton, New York

Automotive Technician Assistant, **June 2024** – **August 2024**

- Supported senior mechanics in performing part replacements, maintenance, and diagnostics on various vehicle systems.

Engineering Extracurriculars

Rice Formula SAE

Suspension Team Lead, **January 2026 – Present**

- Lead a five-person suspension team development from initial geometry definition through component design and analysis.
- Developed tire force and slip-angle models using Formula SAE Tire Test Consortium data to evaluate tire behavior under varying loads and camber conditions to inform suspension design.
- Modeled suspension corner assemblies in SolidWorks and performed FEA on uprights, control arms, and rockers under braking and cornering loads.
- Led subteam design reviews and established the suspension geometry baseline and design documentation for future vehicles.

Rice Wind Energy

Mechanical Design Team, **September 2024 – May 2025**

- Designed and fabricated the base structure for a floating offshore wind turbine, contributing to a second-place national finish.
- Co-developed a blade-pitch control mechanism and applied CAD, FEA, and structural analysis to evaluate mechanical performance.

Skills

CAD & Analysis:
SolidWorks, ANSYS
Vehicle Dynamics: Tire modeling, suspension analysis
Programming:
Python, C++, MATLAB, Swift, Arduino
Fabrication:
CNC milling, CAM, machining, welding, water jet, laser cutting, FDM, SLA, SLS 3D printing, and soldering

Personal Projects

Quadcopter July 2026

- Designed and built a quadcopter from scratch on a $300 budget, 3D-printing the full frame in PA nylon to a 300 g frame mass.
- Sized and integrated motors, ESCs, and flight-control electronics; optimized the frame for stiffness-to-weight.

RC Car June 2026

- Designed and assembled an RC car using salvaged electronics and 3D-printed components.

Water Incubator December 2025

- Designed an Arduino-based temperature-control system for bacterial incubation with modular electronics and 24-hour temperature stability.

Turbine Base December 2024

- Designed a floating wind-turbine base using CAD, FEA, and load-distribution analysis under variable wind conditions.